

Muhammad Daffa' Fisabilillah

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Summary

Mobile Application Developer with strong foundations in Android development, machine learning integration, and real-time computer vision. Experienced in building end-to-end mobile solutions using Kotlin, MVVM architecture, and modern Android libraries including Jetpack, CameraX, and Firebase. Proficient in deploying lightweight on-device AI models (TFLite, ML Kit) and working with custom CNNs and pose estimation pipelines. Demonstrated ability to work in cross-functional teams and lead student-led projects and organizations. Actively contributed to impactful projects ranging from fitness motion tracking to mobile image classification with offline capabilities.

Education

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| Institut Teknologi Sepuluh Nopember , Computer Engineering | Sept 2021 – Sept 2025 |
| • GPA: 3.53/4.0 | |
| MAN Insan Cendekia OKI , Science | Sept 2018 – Sept 2021 |
| • GPA: 92/100 | |

Experience

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| Mobile Android Developer , PT. Adinata Charming Emmanuel – Surabaya, Indonesia (Hybrid) | Jul 2024 – Feb 2025 |
| <ul style="list-style-type: none">Developed a membership-based Android application connecting users with partnered merchants using Kotlin and Clean Architecture (MVVM) for high scalability and maintainabilityImplemented secure authentication flow (registration, login, OTP/email verification, and token refresh) with full API integrationBuilt UI components in XML following UI/UX design specs, adding animations and smooth transitions for enhanced user experienceHandled core promo flows: promo creation, activation, referral redemption, and promo history displayCollaborated with backend engineers to ensure seamless integration with RESTful APIs (Auth, Token, Promo, and Merchant modules) | |
| Mobile Application Developer (Internship) , Bangkit Academy – Remote (Indonesia) | Feb 2024 – Aug 2024 |
| <ul style="list-style-type: none">Completed a 20-week career readiness program led by Google, GoTo, and Traveloka, focused on industry-level mobile developmentBuilt Android applications using Kotlin, Jetpack Compose, and MVVM, integrating REST APIs and Room local databaseIntegrated on-device Machine Learning features using TensorFlow Lite for real-time predictionContributed to a cross-functional Capstone Project team (ML, Cloud, Mobile) solving real-world problemsParticipated in mentorship sessions, weekly code reviews, and career development workshops | |

Projects

BAKI: Sport Motion Detector Based on ML Kit in Mobile App

- Designed an Android fitness app offering real-time posture correction and motion analysis using ML Kit Pose Detection API and KNN classifier
- Built custom motion classifier on 33 3D skeletal points with CSV dataset and Exponential Moving Average smoothing
- Implemented full CameraX-based live pose tracking, Material Design UI/UX, and Clean Architecture using Dagger Hilt

- Included features: OTP-based authentication, performance dashboard, repetition counter, reflex and anticipation training
- Tools Used: Kotlin, ML Kit, CameraX, Dagger Hilt, Jetpack Navigation, Firebase, CSV Parsing

DogVision: On-Device Dog Breed Classifier Using PyTorch and TFLite

- Developed a custom CNN with residual blocks and Mixup data augmentation to classify 10 dog breeds with high accuracy
- Optimized model for mobile deployment: PyTorch → ONNX → TensorFlow → TFLite
- Built Android app with full offline capability, real-time image classification, and Room-based local history tracking
- Tools Used: PyTorch, ONNX, TensorFlow Lite, Kotlin, Room, Android Studio

Digital Twin for Virtual Music Concert Using Metahuman UE5

- Developed a photorealistic avatar for a virtual musician using Unreal Engine 5 and Metahuman, enabling immersive concert performance in a 3D environment
- Integrated motion capture systems (Move AI, Rokoko, DeepMotion), facial tracking (Live Link Face, Face Landmarker Link 02), and clothing simulation (Marvelous Designer)
- Utilized HyperHuman AI and Polycam for detailed facial mesh generation and accurate identity mapping
- Achieved natural motion and facial expression synchronization with real-time rendering performance at 1080p 30 FPS
- Evaluated avatar realism and blendshape precision using custom analysis pipeline with MediaPipe for eye and mouth landmarks

Mini Game for Playing Card Classification using Deep Learning

- Built a terminal-based game with webcam card recognition using custom CNN model and real-time prediction with OpenCV
- Implemented preprocessing pipeline: thresholding, perspective transform, and label prediction using Keras
- Supported real-time classification, data capture, and interactive gameplay fully within command line
- Tools Used: Python, Keras, OpenCV, NumPy

Organizations

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| Human Resources Development Staff, HIMATEKKOM – ITS | Apr 2023 - Mar 2024 |
| <ul style="list-style-type: none"> • Coordinated internal development programs and talent empowerment initiatives in the Computer Engineering Student Association • Supported recruitment, performance evaluation, and team-building activities to enhance member engagement | |
| Head of Event Organizer Subdivision, MAGE 9 – ITS | Feb 2023 – Nov 2023 |
| <ul style="list-style-type: none"> • Led scheduling and coordination of multiple event categories including competitions, workshops, and webinars • Oversaw preparation, logistics, and cross-functional collaboration across all event teams to ensure seamless execution | |
| Staff of Computer Olympiad Division, MAGE 8 – ITS | Jun 2022 – Dec 2022 |
| <ul style="list-style-type: none"> • Designed olympiad technical guidelines and selected problem sets in collaboration with academic advisors • Managed competition logistics and ensured adherence to academic and technical standards | |

Technologies

Languages: Kotlin, Java, Python, XML, SQL

Frameworks & Libraries: Jetpack Compose, Android Jetpack (Navigation, Room, ViewModel, LiveData), Dagger Hilt, PyTorch, TensorFlow Lite, OpenCV, Keras

Tools & Platforms: Android Studio, Firebase, Git, CameraX, ML Kit, ONNX, CSV Processing, Gradle, GitHub, Jupyter Notebook

Concepts: Clean Architecture (MVVM), Agile Development (Scrum), On-Device Machine Learning, Real-Time Image Processing, Pose Estimation, Mobile Deployment Optimization, REST API Integration, Material Design